

GMX1000 Sound Engine Register Document

2005/03/10

1.Register Map

0	General parameter
020H	Channel status parameter
040H	Wave memory access
200H	ESP parameter
400H	WMA parameter
800H	TSP high parameter
C00H	TSP low parameter
FFFH	

2-1 General parameter

Addr	Register	R/W	15/7	14/6	13/5	12/4	11/3	10/2	9/1	8/0
000	SYSTM	W	CHTST	WMBIH	PCEBE	RXES	RXEN	TXEN	EMCL	START
001			—							
002	MODE	W	SRNE	EBBRS[1:0]		CHLV[2:0]			SOBWS[1:0]	
003			—							
002	ITPR	W	ITPR[7:0]							
003			—							

[SYSTEM REGISTER] : (RESET 0000H)

START: Start Sound Generator and Sound Effect

0: Stop 1: Start

EMCL: Initialize Sound Effect RAM (Fill Zero)

0: Off 1: On

TXEN: Turn on the output to Voice Recorder

0: Off 1: On

RXEN: Turn on the input of Voice Recorder or PCM data

0: Off 1: On

RXEN: Enable Effect Send of PCM data

0: Disable 1: Enable

PCEBE: Set Pan of PCM data (4CH)

0: Front 1: Back

WMBIH: Use wave memory bus

0: Enable 1: Disable

CHTST: Set Test mode

0: Normal mode 1: Test mode

[MODE REGISTER]

SOBWS[1:0]: Set bit width of the serial data of D/A converter

0: 16bits 1: 18bits 2: 20bits 3: 24bits

CHLV[2:0]: Set output level of Sound Engine Channel

0: 1/128 1: 1/64 2: 1/32 3: 1/16 4: 1/8 5: 1/4 6: 1/2 7: 1

EBBRS[1:0]: Set lower bit rounding off handling of conversion from data bus 20 bits of effect department to memory data 16 bits

0: 16bits cutting off 1: 20bits rounding off 2: 18bits rounding off 3: 17bits rounding off

SRNE: Set Surround of Reverb (4CH)

0: Off 1: On

[ITPR REGISTER]

ITPR[7:0]: Value of Envelope-Interpolate

2-2 Channel Status Parameter

Addr	Register	R/W	15/7	14/6	13/5	12/4	11/3	10/2	9/1	8/0
020	CHST	R/W	CH7	CH6	CH5	CH4	CH3	CH2	CH1	CH0
021			CH15	CH14	CH13	CH12	CH11	CH10	CH9	CH8
022			CH23	CH22	CH21	CH20	CH19	CH18	CH17	CH16
023			CH31	CH30	CH29	CH28	CH27	CH26	CH25	CH24
024			CH39	CH38	CH37	CH36	CH35	CH34	CH33	CH32
025			CH47	CH46	CH45	CH44	CH43	CH42	CH41	CH40
026			CH55	CH54	CH53	CH52	CH51	CH50	CH49	CH48
027			CH63	CH62	CH61	CH60	CH59	CH58	CH57	CH56
028	SCHST		CH7	CH6	CH5	CH4	CH3	XH2	CH1	CH0
029			CH15	CH14	CH13	CH12	CH11	CH10	CH9	CH8
02A			CH23	CH22	CH21	CH20	CH19	CH18	CH17	CH16
02B			CH31	CH30	CH29	CH28	CH27	CH26	CH25	CH24
02C			CH39	CH38	CH37	CH36	CH35	CH34	CH33	CH32
02D			CH47	CH46	CH45	CH44	CH43	CH42	CH41	CH40
02E			CH55	CH54	CH53	CH52	CH51	CH50	CH49	CH48
02F			CH63	CH62	CH61	CH60	CH59	CH58	CH57	CH56
030	SCHON	W	—							
031			—							

[CHST] : CHANNEL STATE REGISTER

Sound Engine Channel On/Off (Read / Write)

Each bit supports Sound Engine Channel

0: Off 1: On

*This register becomes off by wave mode setting automatically

* When assign channel, refer to this bit

[SCHST] : SYNCRONUS CHANNEL STATE REGISTER

The register who sets a channel doing synchronization on.

Use it with the SCHON register

0: Disable 1: Enable

[SCHON] : SYNCRONUS CHANNEL ON REGISTER

SCHST Enable Channel Synchronization on. WriteData :don't care.

CHST becomes On a channel of Enable SCHST

SCHST becomes All Zero(Disable)

When there is the tone group which you want to turn on at the same time, use this

2-3 Wave Memory Access

Addr	Register	R/W	15/7	14/6	13/5	12/4	11/3	10/2	9/1	8/0
040	WMAL	W	WMADR[23:16]							
041			—							
042	WMAH	W	WMADR[7:06]							
043			WMADR[15:8]							
044	WMDT	R/W	WMDATA[7:0]							
045			—							

*This register can access only the whole sound engine stop

[WMAL] : WAVE MEMORY ADDRESS LOW REGISTER

WMADR[15:08] : Wave memory address (Lower 16 bits)

[WMAH] : WAVE MEMORY ADDRESS HIGHT REGISTER

WMADR[23:16] : Wave memory address (Higher 8 bits)

[WMDT] : WAVE MEMORY DATA REGISTER

WMDATA[7:0] : Wave memory data

*Wave Data Write --- 1.WMAL(W) 2.WMAH(W) 3.WMDAT(W)

*Wave Data Read --- 1.WMAL(W) 2.WMAH(W) 3.WMDAT(R)

2-4 WMA parameter

ADDRESS + (800H+16×CHNO)

Addr	Register	R/W	15/7	14/6	13/5	12/4	11/3	10/2	9/1	8/0
+0	PIT	W	*PITACCH[7:0]							
+1	ACCH		*PITACCH[15:8]							
+2	INIT	W	—	PDIR	PLCD	PWA	LCD[3:0]			
+3										
+4	MODE	W	FMOD	WMOD	LMOD[1:0]		—		PITCH[17:16}	
+5			WCS[3:0]					—		
+6	WAOF	W	WAOF[7:0]							
+7			WAOF[15:8]							
+8	LP1	W	LP1[7:0]							
+9			LP1[15:8]							
+A	LP2	W	LP2[23:16]							
+B			LP2[23:16]							
+C	PITCH	W	PITCH[7:0]							
+D			PITCH[15:8]							
+E	PIT	W	*PITACCL[7:0]							
+F	ACCL		*PITACCL[15:8]							

[INIT] : INITIAL WAVE DATA REGISTER

LCD[3:0] : Level data of previous wave(ADPCM). Write Wave Parameter

PWA : Wave address(LSB) of previous wave. Write 0

PLCD : Level control bit of previous wave(ADPCM). Write 0

PDIR : Course to advance of previous wave. Write 0

[MODE] : WAVE MODE REGISTER

LMOD[1:0] : Wave Loop Mode

0: Normal 1: Alternate1 2: One shot 3: Alternate2

One shot:When wave position arrives at LoopPoint1, CHST becomes 0, and Sound Engine Channel stops

WMOD : Wave Mode

0: ADPCM8bits Wave 1: Linear8bits Wave

FMOD:Set Filter Mode

0: Low pass Filter 1: High pass Filter

[WAOF]:WAVE ADDRESS OFFSET

WAOF[15:0]:Wave Address Offset

Wave Address = PITACCH[15:0] + WAOF[15:0] * 16

[LP1] : LOOP POINT1 REGISTER

LP1[15:0]: Address of LoopPoint1

[LP2] : LOOP POINT 2 REGISTER

LP2[15:0] : Address of LoopPoint2

[PITCH] : PITCH REGISTER

PITCH[17:00] : Wave Pitch (linear data)

[PITACCL] : PITCH ACCUMLATOR LOW REGISTER

PITACCL[15:0] Pitch Accumulator (Lower 16 bits) Write Zero

[PITACCH] : PITCH ACCULATOR HIGHT REGISTER

PITACCH[15:0] Pitch Accumulator (Higher 16 bits) Write Zero

2-5 TSP parameter

ADDRESS+(800H+16×CHNO)

Addr	Register	R/W	15/7	14/6	13/5	12/4	11/3	10/2	9/1	8/0
+0	WDAT	W	*WAVE DATA[19:12]							
+1			—							
+2	FILT	W	*FILT ACC1[19:12]							
+3	ACC1		—							
+4	FILT	W	*FILT ACC2[19:12]							
+5	ACC2		—							
+6	FILT	W	*FILT FRQ1 [7:0]							
+7	FRQ 1		—							
+8	FILT	W	F1LT FRQ2[7:0]							
+9	FRQ2		—							
+A	AMP	W	AMP LEV[7:0]							
+B	LEV		—							
+C	PAN L	W	PAN L[7:0]							
+D			—							
+E	REV	W	REV SDL							
+F	SDL		—							

ADDRESS+(C00H+16×CNNO)

Addr	Register	R/W	15/7	14/6	13/5	12/4	11/3	10/2	9/1	8/0
+0	WDAT	W	*WAVE DATA[7:0]							
+1			—				*WAVE DATA[11:8]			
+2	FILT1	W	*FILT ACC1[7:0]							
+3	ACC1		—				*FILT ACC1[11:8]			
+4	FILT	W	*FILT ACC2[7:0]							
+5	ACC2		—				*FILT ACC2[11:8]			
+6	FRQ	W	*FLT FRQ1 ACC1[7:0]							
+7	ACC1		—				*FLT FRQ1 ACC[11:8]			
+8	FRQ	W	*FLT FRQ2 ACC2[7:0]							
+9	ACC2		—				*FLT FRQ2 ACC2[11:8]			
+A	LEV	W	*AMP LEV ACC[7:0]							
+B	ACC		—				*AMP LEV ACC[11:8]			
+C	PANL	W	PANR[3:0]				DLY SDL[3:0]			
+D			—				PANR[7:4]			
+E	CHO	W	CHO SDL[3:0]				DLY SDL[7:4]			
+F	SDL		—				CHO SDL[7:4]			

WAVE DATA[19:0] : Previous Wave Data. Write Zero
FILT ACC1[19:0] : Filter Accumulator1. Write Zero
FILT ACC2[19:00] : Filter Accumulator2. Write Zero
FILT FRQ1[7:0] : Filter Frequency1. Write(80H—7FH)
FILT FRQ2[7:0] : Filter Frequency2. Write(80H—7FH)
FILT FRQ1 ACC[11:0] : Filter Frequency1 Accumulator. Write = FILT FRQ1 * 16.
FILT FRQ2 ACC[11:0] : Filter Frequency2 Accumulator. Write = FILT FRQ2 * 16.
AMP LEV[7:0] : Amplitude Level. Write(80H—7FH)
AMP LEV ACC[11:0] : Amplitude Level Accumulator. Write = AMP LEV * 16.
PANL[7:0] : Pan Left Level
PANR[7:0] : Pan Right Level
REV SDL[7:0] : Reverb Send Level
CHO SDL[7:0] : Chorus Send Level
DLY SDL[7:0] : Delay Send Level

2-6 Sound Effect parameter

Addr	Register	R/W	D[15:8]	D[7:0]
200	REV1 ADR	W		REV1ADR0[15:0]
202				REV1ADR1[15:0]
204				REV1ADR2[15:0]
206				REV1ADR3[15:0]
208				REV1ADR4[15:0]
20A				REV1ADR05[15:0]
20C				REV1ADR6[15:0]
20E				REV1ADR7[15:0]
210				REV1ADR8[15:0]
212				REV1ADR9[15:0]
214				REV1ADR10[15:0]
216				REV1ADR11[15:0]
218				REV1ADR12[15:0]
21A				REV1ADR13[15:0]
21C				REV1ADR14[15:0]
21E				REV1ADR15[15:0]
220				REV1ADR16[15:0]
222				REV1ADR17[15:0]
224				REV1ADR18[15:0]
226				REV1ADR19[15:0]
228				REV1ADR20[15:0]
22A				REV1ADR21[15:0]
22C				REV1ADR22[15:0]
22E				REV1ADR23[15:0]
230				REV1ADR24[15:0]
232				REV1ADR25[15:0]
234				REV1ADR26[15:0]
236				REV1ADR27[15:0]
238				REV1ADR28[15:0]
23A				REV1ADR29[15:0]
23C				REV1ADR30[15:0]
23E				REV1ADR31[15:0]
280	REV2	W		REV2ADR0[15:0]
282				REV2ADR1[15:0]
284				REV2ADR2[15:0]
286				REV2ADR3[15:0]
288				REV2ADR4[15:0]
28A				REV2ADR5[15:0]
28C				REV2ADR6[15:0]
28E				REV2ADR7[15:0]
290				REV2ADR8[15:0]
292				REV2ADR9[15:0]
294				REV2ADR10[15:0]
296				REV2ADR11[15:0]
298				REV2ADR12[15:0]
29A				REV2ADR13[15:0]
29C				REV2ADR14[15:0]
29E				REV2ADR15[15:0]
2A0				REV2ADR16[15:0]

2A2	ADR		REV2ADR17[15:0]	
2A4			REV2ADR18[15:0]	
2A6			REV2ADR19[15:0]	
2A8			REV2ADR20[15:0]	
2AA			REV2ADR21[11:0]	
2AC			REV2ADR22[15:0]	
2AE			REV2ADR23[15:0]	
2B0			REV2ADR24[15:0]	
2B2			REV2ADR25[15:0]	
2B4			REV2ADR26[15:0]	
2B6			REV2ADR27[15:0]	
2B8			REV2ADR28[15:0]	
2BA			REV2ADR29[15:0]	
2BC			REV2ADR30[15:0]	
2BE			REVADR31[15:0]	
300		W		PCM LEV[7:0]
302		W	MIC2 LEV[7:0]	MIC1 LEV[7:0]
304		W	EQ1 BD0 LEV[7:0]	EQ1 BD0 FRQ[7:0]
306		W	EQ1 BD1 LEV[7:0]	EQ1 BD1 FRQ[7:0]
308		W	EQ1 BD2 LEV[7:0]	EQ1 BD2 FRQ[7:0]
30A		W	EQ1 BD3 LEV[7:0]	EQ1 BD3 FRQ[7:0]
30C		W	EQ2 BD0 LEV[7:0]	EQ2 BD0 FRQ[7:0]
30E		W	EQ2 BD1 LEV[7:0]	EQ2 BD1 FRQ[7:0]
310		W	EQ2 BD2 LEV[7:0]	EQ2 BD2 FRQ[7:0]
312		W	EQ2 BD3 LEV[7:0]	EQ2 BD3 FRQ[7:0]
314		W	MIC1 PAN FR[7:0]	MIC1 DIR LEV[7:0]
316		W	MIC2 DIR LEV[7:0]	MIC1 PAN FL[7:0]
318		W	MIC2 PAN FL[7:0]	MIC2 PAN FR[7:0]
31A		W	MIC2 CHO SDL[7:0]	MIC1 CHO SDL[7:0]
31C		W	CHO LPF FRQ[7:0]	PCM CHO SDL[7:0]
31E		W	CHO LFO RATE[15:0]	
320		W	*CHO LFO ACC[15:0]	
322		W	CHO DLY TIME[15:0]	
324		W	CHO OUT LEV[7:0]	CHO MOD DEP[7:0]
326		W	CHO DLY SDL[7:0]	CHO REV1 SDL[7:0]
328		W		CHO FBK LEV[7:0]
32A		W	MIC2 PCH SDL[7:0]	MIC1 PCH SDL[7:0]
32C		W	PCH LFO RATE[15:0]	
32E		W	*PCH LFO ACC[15:0]	
330		W	PCH OFS ADR[15:0]	
332		W	PCH OUT LEV[7:0]	PCH MOD DEP[7:0]
334		W	PCH PAN FL[7:0]	PCH PAN FR[7:0]
336		W	DLY LPF FRQ[7:0]	PCH LPF FRQ[7:0]
338		W	DLY C LEV[7:0]	DLY FBK FRQ[7:0]
33A		W	DLY L LEV[7:0]	DLY R LEV[7:0]
33C		W	DLY REV1 SDL[7:0]	DLY OUT LEV[7:0]
33E		W	MIC2 REV1 SDL[7:0]	MIC1 REV1 SDL[7:0]
340		W	REV1 PLF FRQ[7:0]	PCM REV1 SDL[7:0]
342		W	REV1 DLY0 LEV[7:0]	REV1 FBK0 LEV[7:0]
344		W	—	REV1 BYP0 LEV[7:0]
346		W	REV1 DLY1 LEV[7:0]	REV1 FBK1 LEV[7:0]
348		W	REV1 ALPF FRQ	REV1 BYP1 LEV[7:0]
34A		W		REV1 REV TIME[7:0]

34C		W	REV1 FL0 LEV[7:0]	REV1 FR0 LEV[7:0]
34E		W	REV1 FL1 LEV[7:0]	REV1 FR1 LEV[7:0]
350		W	REV1 FL2 LEV[7:0]	REV1 FR2 LEV[7:0]
352		W	REV1 FL3 LEV[7:0]	REV1 FR3 LEV[7:0]
354		W	REV1 FL OUT L[7:0]	REV1 FR OUT LEV[7:0]
356		W	REV1 BL0 LEV[7:0]	REV1 BR0 LEV[7:0]
358		W	REV1 BL1 LEV[7:0]	REV1 BR1 LEV[7:0]
35A		W	REV1 BL2 LEV[7:0]	REV1 BR2 LEV[7:0]
35C		W	REV1 BL3 LEV[7:0]	REV1 BR3 LEV[7:0]
35E		W	REV1 BL OUT LEV[7:0]	REV1 BR OUT LEV[7:0]
360		W	MIC2 REV2 SDL[7:0]	MIC1 REV2 SDL[7:0]
362		W	REV2 PLPF FRQ[7:0]	PCH REV2 SDL[7:0]
364		W	REV2 DLY0 LEV[7:0]	REV2 FBK0 LEV[7:0]
366		W	—	REV2 BYP0 LEV[7:0]
368		W	REV2 DLY1 LEV[7:0]	REV2 FBK1 LEV[7:0]
36A		W	REV2 ALPF FRQ	REV2 BYP1 LEV[7:0]
36C		W		REV2 TIME2 LEV[7:0]
36E		W	REV2 FL0 LEV[7:0]	REV2 FR0 LEV[7:0]
370		W	REV2 FL1 LEV[7:0]	REV2 FR1 LEV[7:0]
372		W	REV2 FL2 LEV[7:0]	REV2 FR2 LEV[7:0]
374		W	REV2 FL3 LEV[7:0]	REV2 FR3 LEV[7:0]
376		W	REV2 FL OUT LEV[7:0]	REV2 FR OUT LEV[7:0]
378		W	REV2 BL0 LEV[7:0]	REV2 BR0 LEV[7:0]
37A		W	REV2 BL1 LEV[7:0]	REV2 BR1 LEV[7:0]
37C		W	REV2 BL2 LEV[7:0]	REV2 BR2 LEV[7:0]
37E		W	REV2 BL3 LEV[7:0]	REV2 BR3 LEV[7:0]
380		W	REV2 BL OUT LEV[7:0]	REV2 BR OUT LEV[7:0]
382		W		EQ GAIN[7:0]
384		W	EQ3 BD0 LEV[7:0]	EQ3 BD0 FRQ[7:0]
386		W	EQ3 BD1 LEV[7:0]	EQ3 BD1 FRQ[7:0]
388		W	EQ3 BD2 LEV[7:0]	EQ3 BD2 FRQ[7:0]
38A		W	EQ3 BD3 LEV[7:0]	EQ3 BD3 FRQ[7:0]
3C0		W		
3C2		W		
3C4		W		
3C6		W		
3C8		W	CHO WR ADR[15:0]	
3CA		W		
3CC		W		
3CE		W		
3D0		W		
3D2		W	PCH WR ADR[15:0]	
3D4		W	DLY C ADR[15:0]	
3D6		W	DLY WR ADR[15:0]	
3D8		W	DLY R ADR[15:0]	
3DA		W	DLY L ADR[15:0]	
3DC		W		
3DE		W		

REV1 ADn[15:0] : Reverb1- Delay address (for Sound Generator)
 REV2 ADn[15:0] : Reverb2- Delay address (for Mic)
 PCM LEV[7:0] : Voice recorder / PCM – Level (LR commonness)
 MIC1 LEV[7:0] : Mic1- Level (EQ input)
 MIC2 LEV[7:0] : Mic2 -Level (EQ input)
 EQ1 BDn FRQ[7:0] : Mic1-EQ_Band(n) Frequency
 EQ1 BDn LEV[7:0] : Mic1-EQ_Band(n) Level
 EQ2 BDn FRQ[7:0] : Mic2-EQ_Band(n) Frequency
 EQ2 BDn LEV[7:0] : Mic2-EQ_Band(n) Level
 MIC1 DIR LEV[7:0] : Mic1-Direct Output Level
 MIC1 PAN FR[7:0] : Mic1- Front Right Level
 MIC1 PAN FL[7:0] : Mic1- Front Left Level
 MIC2 PAN FR[7:0] : Mic2- Front Right Level
 MIC2 PAN FL[7:0] : Mic2 -Front Left Level
 MIC1 CHO SDL[7:0] : Mic1-ChorusSendLevel
 MIC2 CHO SDL[7:0] : Mic2- ChorusSendLevel
 PCM CHO SDL[7:0] : PCM/Voice recorder - ChorusSendLevel
 CHO LPF FRQ[7:0] : Chorus-PreLowpassFilter Frequency
 CHO LFO RATE[15:0] : Chorus-LFO(for Modulation) Frequency
 CHO LFO ACC[15:0] : Chorus- LFO(for Modulation) Accumulator. Initialize Zero
 CHO DLY TIME[15:0] : Chorus-Modulation Delay Time
 CHO MOD DEP[7:0] : Chorus-Modulation Depth
 CHO OUT LEV[7:0] : Chorus-Output Level
 CHO REV1 SDL[7:0] : Chorus-Reverb1SendLevel
 CHO DLY SDL[7:0] : Chorus-DelaySendLevel
 CHO FBK LEV[7:0] : Chorus-FeedbackLevel
 MIC1 PCH SDL[7:0] : Mic1-PicthChangerSendLevel
 MIC2 PCH SDL[7:0] : Mic2-PicthChangerSendLevel
 PCH LFO RATE[7:0] : Pitch Changer- LFO(for Modulation) Frequency
 PCH LFO ACC[7:0] : Pitch Changer- LFO(for Modulation) Accumulator. Initialize Zero
 PCH OFS ADn[7:0] : Pitch Changer- LFO(for Modulation)Offset Address
 PCH MOD DEP[7:0] : Pitch Changer-Modulation Depth
 PCH OUT LEV[7:0] : Pitch Changer-Output Level
 PCH PAN R[7:0] : Pitch Changer-FrontRight Level
 PCH PAN L[7:0] : Pitch Changer- FrontLeft Level

PCH LPF FRQ[7:0] : Pitch Changer-PreLowpassFilter Frequency
DLY LPF FRQ[7:0] : Delay –PreLowpassFilter Frequency
DLY FBK FRQ[7:0] : Delay –Feedback Level
DLY C LEV[7:0] : Delay –Center Level
DLY R LEV[7:0] : Delay - Right Level
DLY L LEV[7:0] : Delay –Left Level
DLY OUT LEV[7:0] : Delay - Output Level
DLY REV1 SDL[7:0] : Delay -Reverb1SendLevel
MIC1 REV1 SDL[7:0] : Mic1- Reverb1SendLevel
MIC2 REV1 SDL[7:0] : Mic2- Reverb1SendLevel
PCM REV1 SDL[7:0] : PCM / VoiceRecorderIn-Reverb1SendLevel
REV1 PLF FRQ[7:0] : Reverb1- PreLowpassFilter Frequency
REV1 FBK0 LEV[7:0] : Reverb1-FeedbackLevel(Stage1)
REV1 DLY0 LEV[7:0] : Reverb1-DelayLevel(Stage1)
REV1 BYP0 LEV[7:0] : Reverb1- Bypass Level (Stage1)
REV1 FBK1 LEV[7:0] : Reverb1- FeedbackLevel(Stage2)
REV1 DLY1 LEV[7:0] : Reverb1-Delay Level(Stage2)
REV1 BYP1 LEV[7:0] : Reverb1- Bypass Level (Stage2)
REV1 ALPF FRQ : Reverb1- LowpassFilter Frequency(Stage2)
REV1 REV TIME[7:0] : Reverb1-ReverbTime(Stage2)
REV1 FRn LEV[7:0] : Reverb1-FrontRightLevel(n)
REV1 FLn LEV[7:0] : Reverb1 -FrontLeftLevel(n)
REV1 FR OUT LEV[7:0] : Reverb1- Front Right Master Level
REV1 FL MST LEV[7:0] : Reverb1- Front Left Master Level
REV1 BRn LEV[7:0] : Reverb1- BackRightLevel(n)
REV1 BLn LEV[7:0] : Reverb1- BackLeftLevel(n)
REV1 BR OUT LEV[7:0] : Reverb1-Back Right Master Level
REV1 BL OUT LEV[7:0] : Reverb1- Back Left Master Level
MIC1 REV2 SDL[7:0] : Mic1-Reverb2SendLevel
MIC2 REV2 SDL[7:0] : Mic2- Reverb2SendLevel
PCH REV2 SDL[7:0] : PitchChanger- Reverb2SendLevel
REV2 PLPF FRQ[7:0] : Reverb2- PreLowpassFilter Frequency
REV2 FBK0 LEV[7:0] : Reverb2- FeedbackLevel(Stage1)
REV2 DLY0 LEV[7:0] : Reverb2- DelayLevel(Stage1)
REV2 BYP0 LEV[7:0] : Reverb2- Bypass Level (Stage1)
REV2 FBK1 LEV[7:0] : Reverb2- FeedbackLevel(Stage2)

REV2 DLY1 LEV[7:0] : Reverb2- DelayLevel(Stage2)
REV2 BYP1 LEV[7:0] : Reverb2- Bypass Level (Stage2)
REV2 ALPF FRQ : Reverb2- LowpassFilter Frequency(Stage2)
REV2 REV TIME[7:0] : Reverb2-ReverbTime(Stage2)
REV2 FRn LEV[7:0] : Reverb2- FrontRightLevel(n)
REV2 FLn LEV[7:0] : Reverb2- FrontLeftLevel(n)
REV2 FR OUT LEV[7:0] : Reverb2- Front Right Master Level
REV2 FL OUT LEV[7:0] : Reverb2- Front Left Master Level
REV2 BRN LEV[7:0] : Reverb2- BackRightLevel(n)
REV2 BLN LEV[7:0] : Reverb2- BackLeftLevel(n)
REV2 BR OUT LEV[7:0] : Reverb2- Back Right Master Level
EQ GAIN[7:0] : Equalizer Gain(main)
EQ3 BDN FRQ[7:0] : Equalizer Frequency(main)
CHO WR ADR[15:0] : Chorus-DataWriteAddress
PCH WR ADR[15:0] : Pitch Changer- DataWriteAddress
DLY C ADR[15:0] : Delay-Center Out Read Address
DLY WR ADR[15:0] : Delay- InputDataWriteAddress
DLY R ADR[15:0] : Delay-Right Out Read Address
DLY L ADR[15:0] : Delay-Left Out Read Address